# Kriya AgNova

## AI-Powered Autonomous Biological Crop Care

**Integrating robotics, agronomic intelligence, and biological crop inputs**

A next-generation agricultural platform that continuously scans fields, diagnoses crop stress, and delivers precision biological treatments.

Presented by
**Kriya Agritech**
(A division of Kriya Biosys)

# Agriculture Is Entering a New Era

- Increasing pest and disease pressure
- Pesticide resistance spreading globally
- Soil degradation and declining fertility
- Stricter regulations on chemical crop protection
- Demand for sustainable farming practices

**Agriculture requires smarter crop care systems.**

# The Core Problem

**Detection**
Crop issues are often discovered too late.

**Monitoring**
Large farms make manual monitoring impractical.

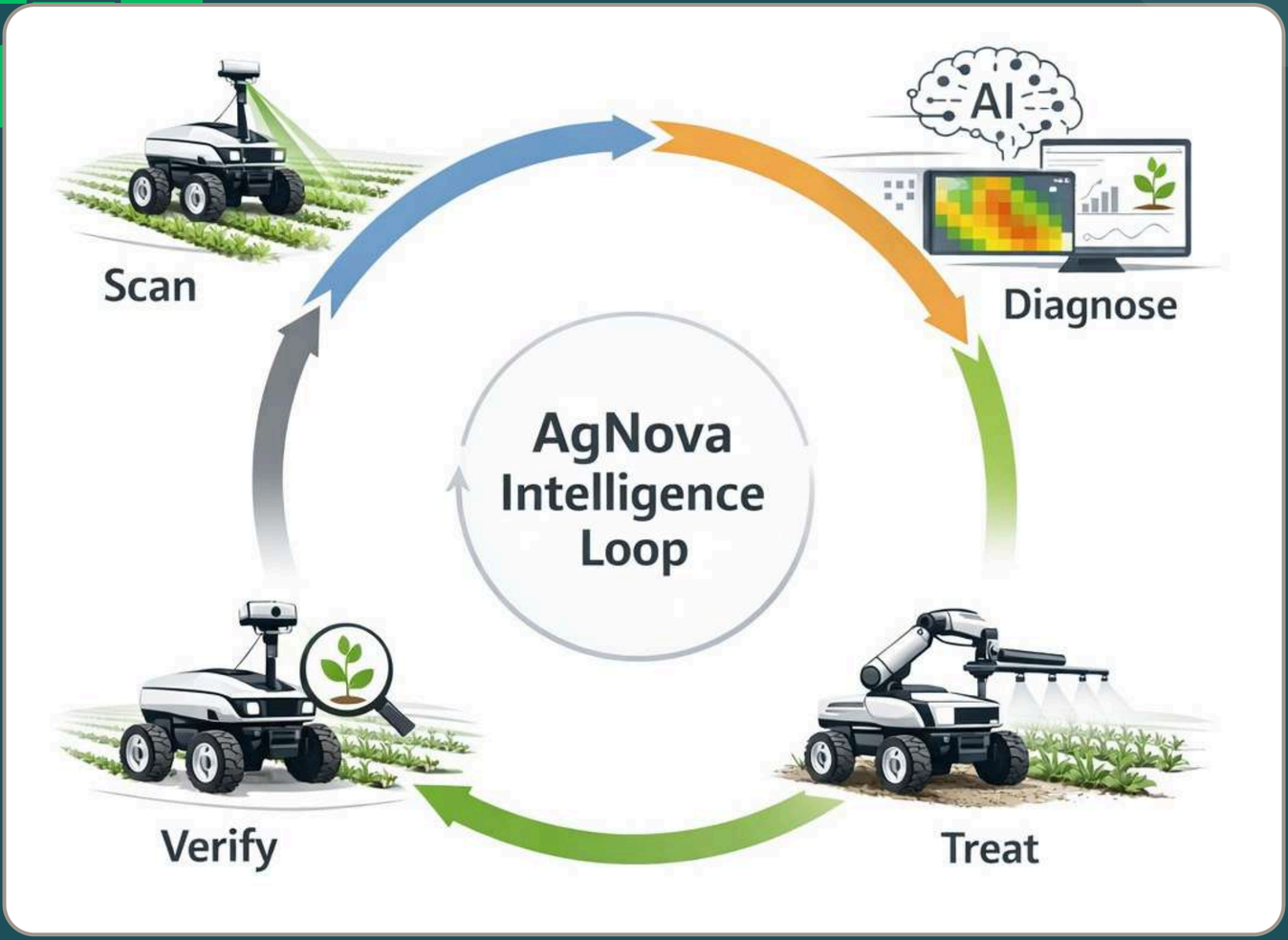
**Treatment**
Most interventions rely on blanket spraying.

# The Precision Agriculture Gap

**But treatment decisions remain manual.**

# Introducing AgNova

An integrated platform combining robotics, AI and biological crop inputs.




**Scan → Diagnose → Treat → Verify**

# Closed Loop Crop Care

### 1 Scan

K-Scout captures field intelligence.

### 3 Treat

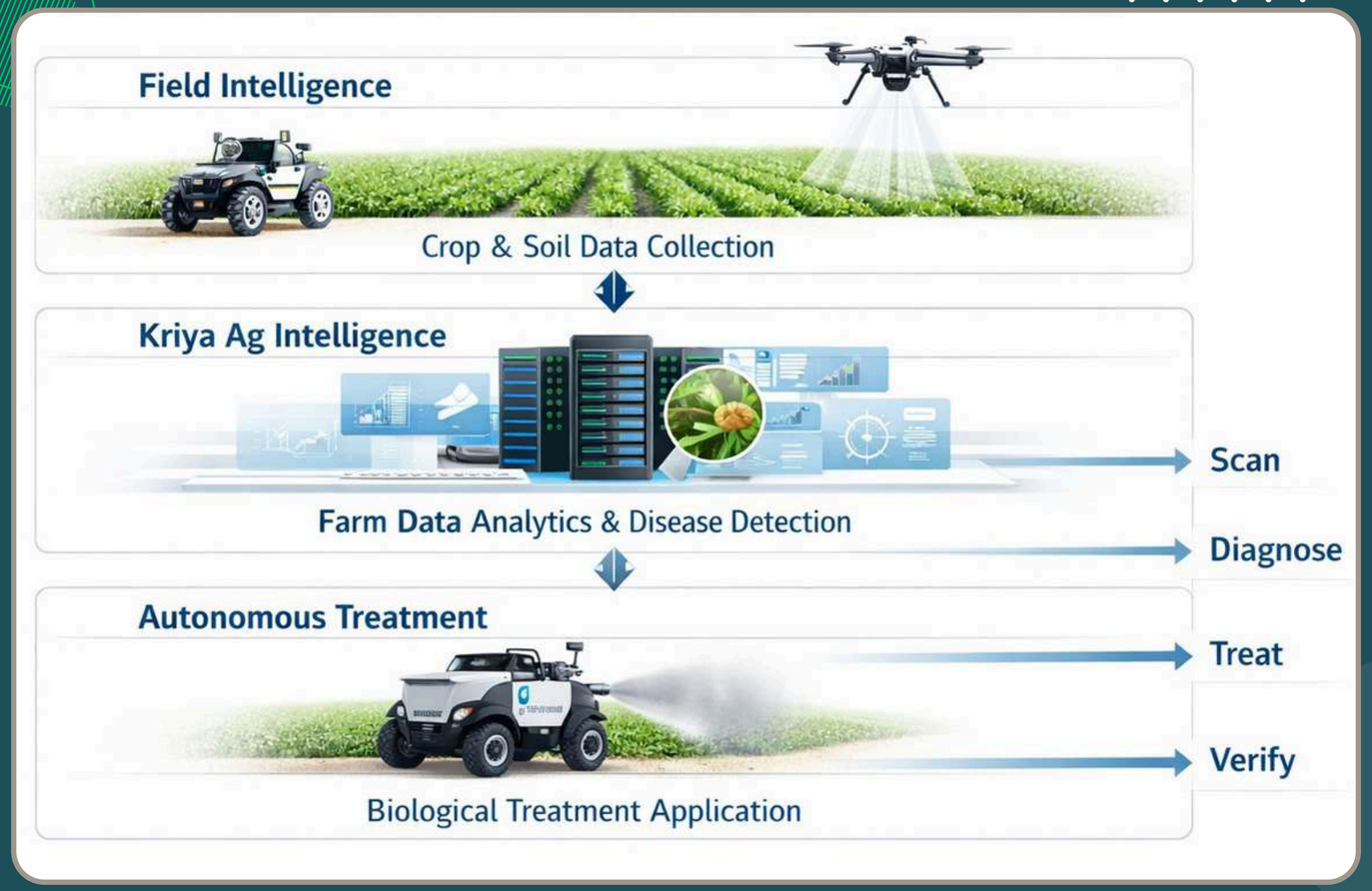
K-Fend delivers targeted treatments

### 2 Diagnose

AI analyzes crop health.

### 4 Verify

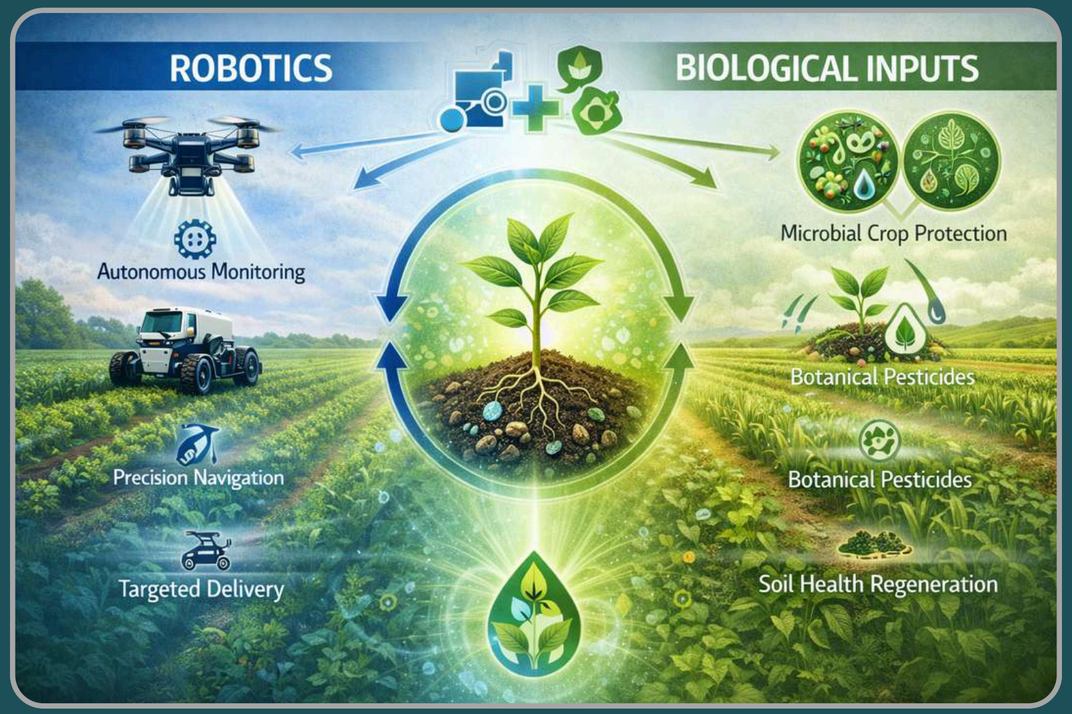
Field data improves the AI.



# Why Robotics + Biologicals

**Robotics**

- Autonomous monitoring
- Precision navigation
- Targeted delivery




**Biological Inputs**

- Microbial crop protection
- Botanical pesticides
- Soil health regeneration

**AgNova operates at the convergence of these technologies.**

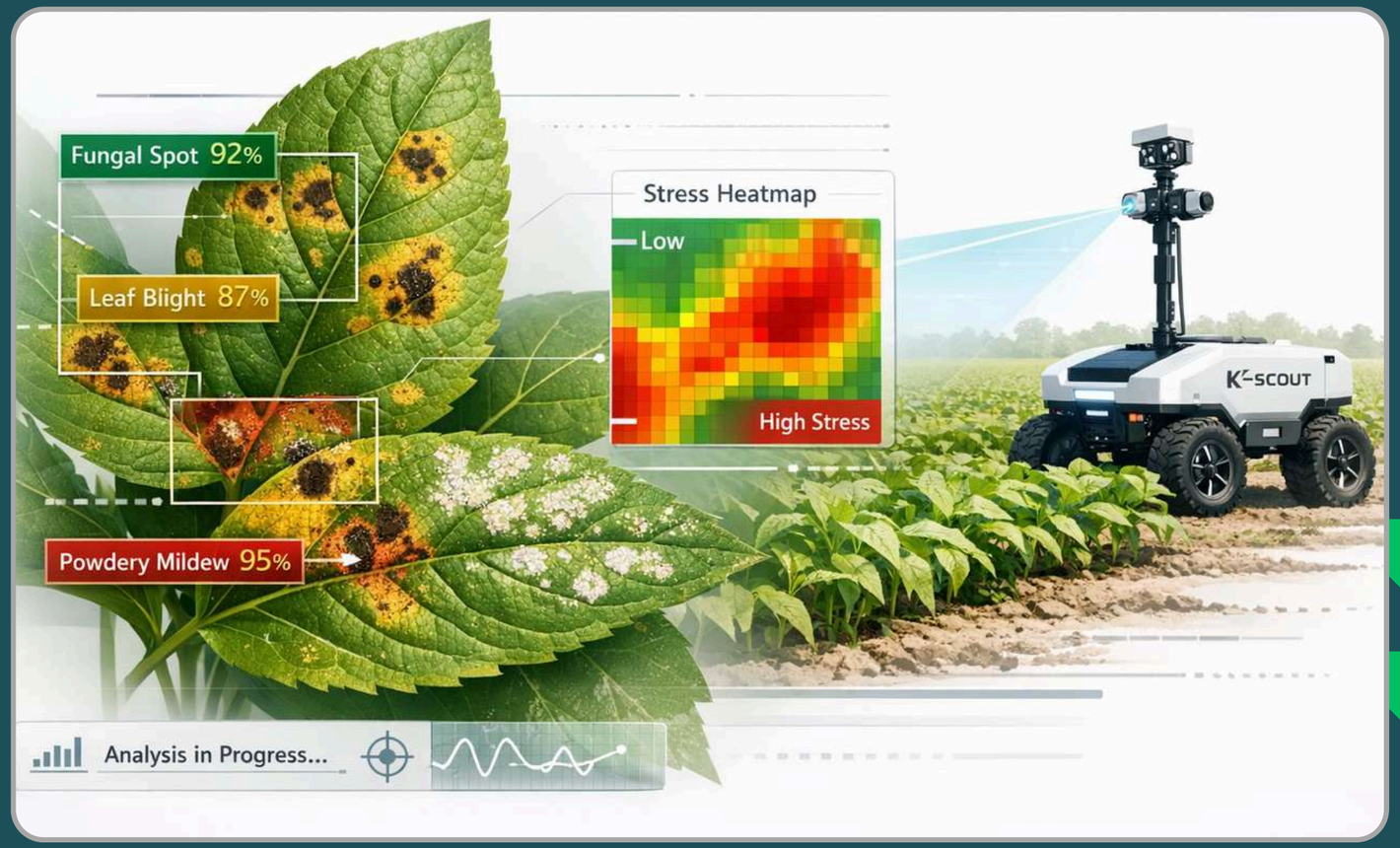
# Platform Architecture

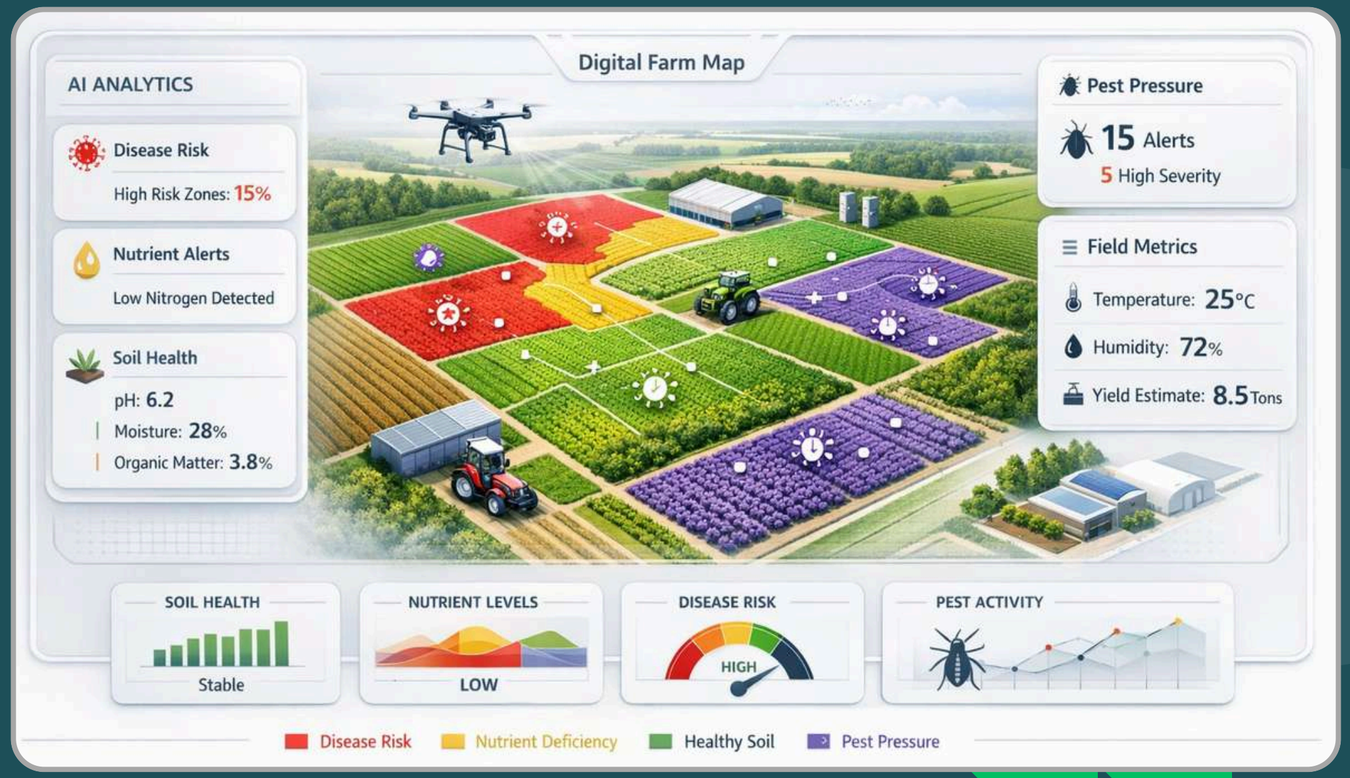
# Operational Workflow

- Continuous field scanning
- AI agronomic analysis
- Precision biological intervention
- Verification and learning

# K-Scout Rover

- RTK GPS navigation
- RGB + multispectral imaging
- Thermal stress detection
- Soil sensing and microclimate monitoring





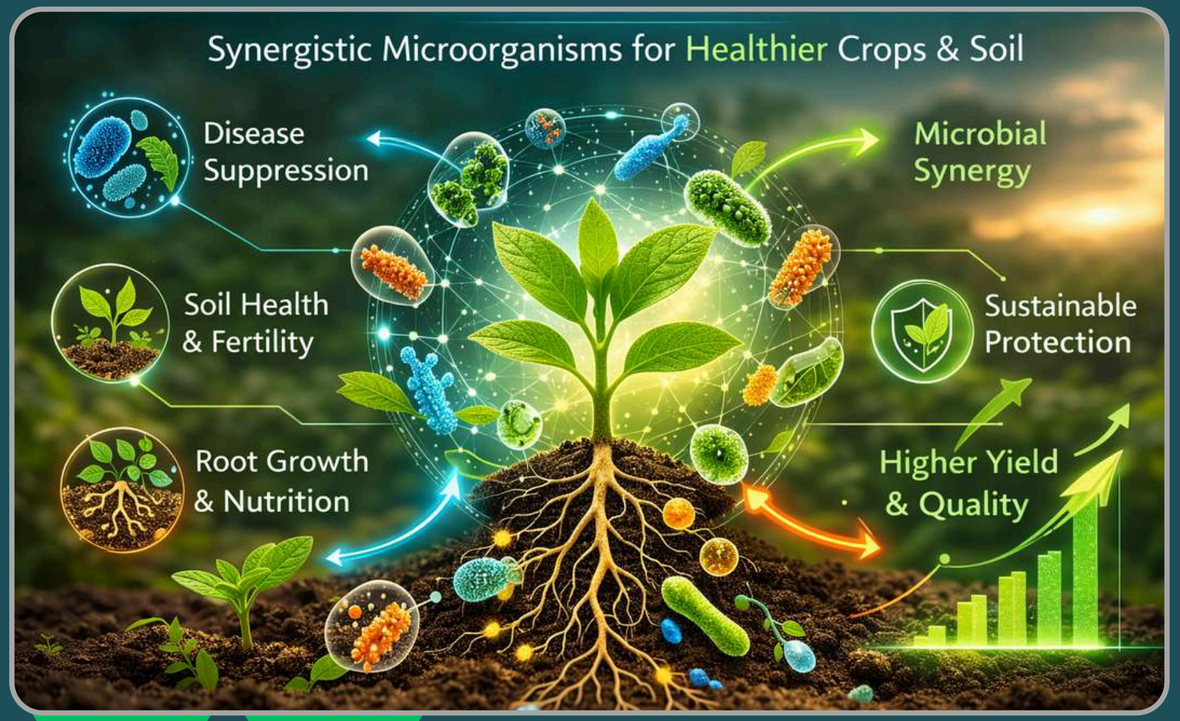

# Kriya Ag Intelligence

- Disease detection models
- Pest pressure analysis
- Nutrient deficiency diagnostics
- Predictive agronomy analytics

# K-Fend Rover

- Precision micro-spraying
- Root-zone microbial injection
- Localized soil drench
- Plant-level treatment targeting

# Dynamic Biological Consortia




**1**

**Microbial Inputs**

- Trichoderma
- Bacillus
- Beauveria

**2**

**Botanical Inputs**

- Neem
- Essential oils
- Plant extracts

**3**

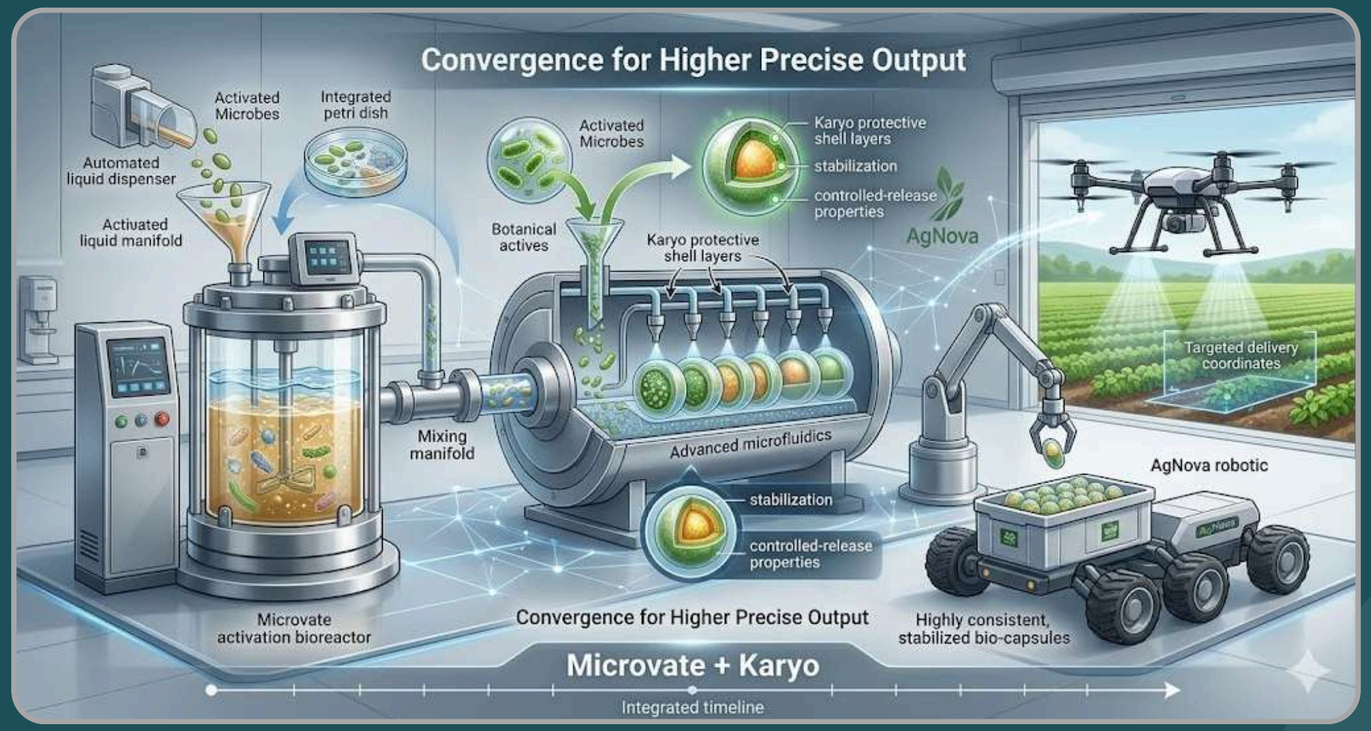
**Biostimulants**

- Seaweed
- Humic substances

# Microvate + Karyo

**Microvate**

Just-in-time microbial activation



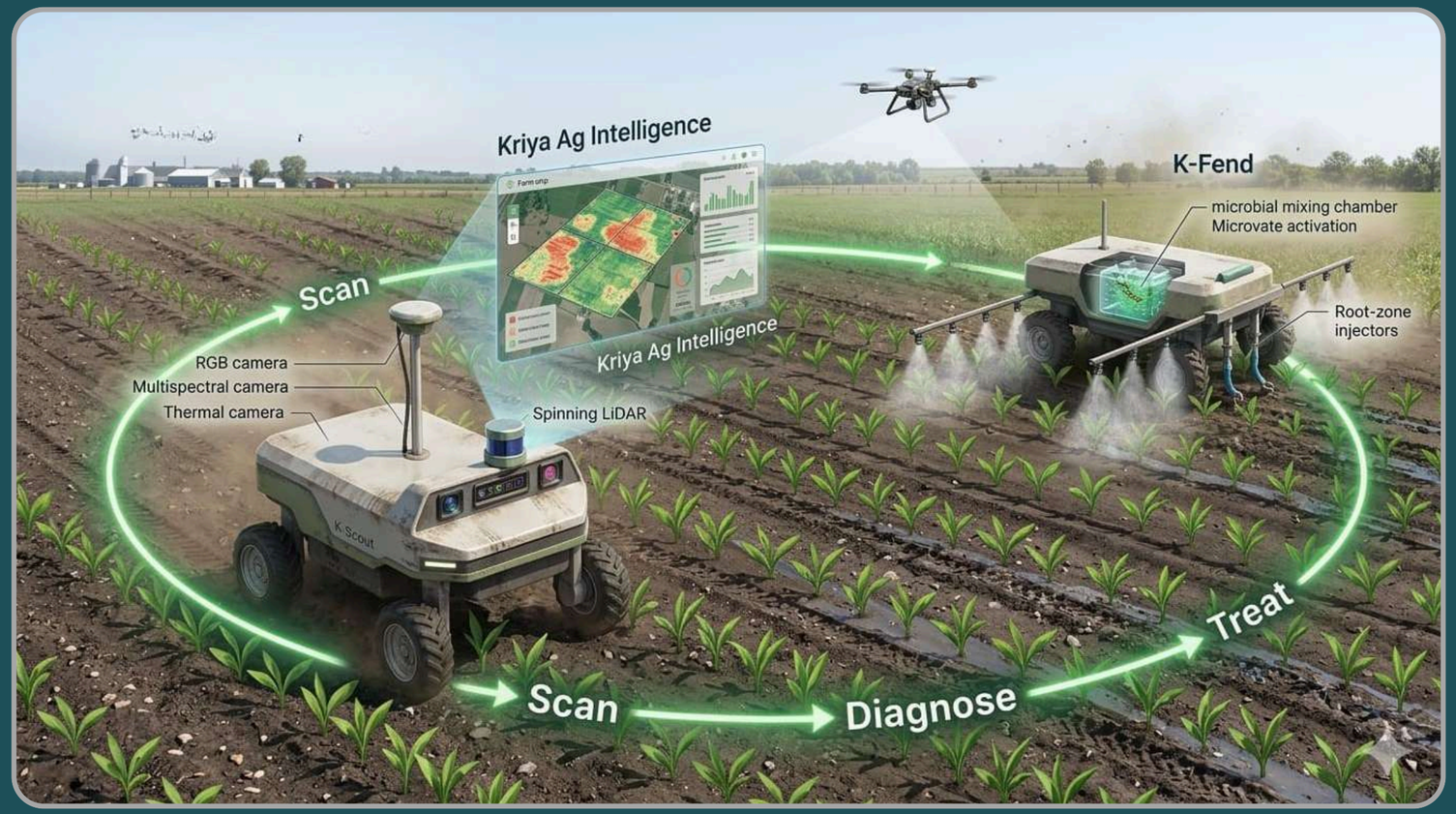

**Karyo**

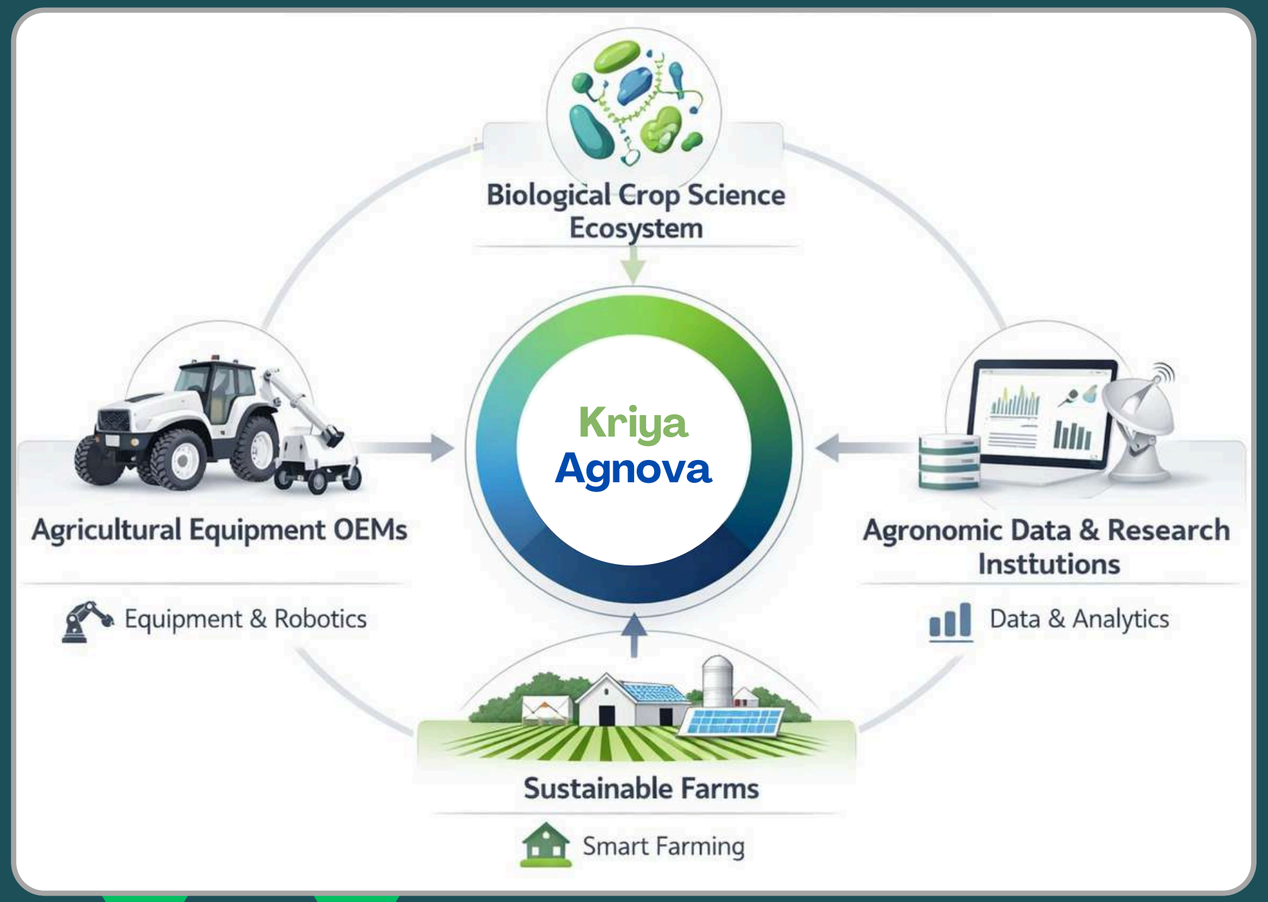
Encapsulation and stabilization technology

**Maximizing biological effectiveness in the field.**

# Digital Farm Twin

- Real-time crop health mapping
- Soil condition monitoring
- Disease risk alerts
- Treatment history tracking






# Technology Moat

1. Robotics Platform
2. Agronomic AI
3. Biological Technologies
4. Agricultural Data

# Strategic Partnerships

- Agricultural equipment OEMs
- Biological crop science ecosystem
- Agronomic research institutions
- Farm deployment partners

# Product Development Roadmap

- Autonomous field intelligence platform
- Agronomic AI expansion
- Precision biological treatment systems
- Integrated robotic crop care
- Autonomous biological agriculture

# The Future of Crop Care

## Kriya AgNova

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